Mitul Tandon

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Research Interests

I am broadly interested in Computer Architecture and Hardware Acceleration for emerging workloads, spanning *microarchitecture design*, *domain-specific accelerators*, and *processing-in-memory architectures*.

Education

Indian Institute of Technology Bombay *(Oct 2022 - Jul 2027)*
Integrated Bachelor and Master of Technology (Dual Degree) in Electrical Engineering *GPA: 8.72/10*

Korea Advanced Institute of Science and Technology (KAIST) *(Sep 2025 - Dec 2025)*
Semester Exchange, Fall 2025

Publications

- M. Tandon, T. Thorawade, V. Venkitaraman, K. Kokkiligadda, V. Singh. **"PM: An Efficient LUT-based Processing-using-Memory Architecture."** *Under review at the IEEE International Symposium on Circuits and Systems* **(ISCAS'27)**
- M. Tandon. **"Accelerating Sparse Matrix Multiplication with a Statically Bound Dataflow."** *Under review at an IEEE computer architecture journal*
- V. Venkitaraman†, M. Tandon†, T. Thorawade, N. Barhate, K. Kokkiligadda, N. Sharma, N. Singh, N. Boran, V. Singh. **"OCEAN: Offset Data-aware Cache Management Scheme for Graph Analytics."** *Accepted as a poster at the IEEE International Conference on High Performance Computing, Data, and Analytics* **(HiPC'26)**
- S. Jeong, C. Ahn, S. Bheemreddy, L. Cooper, K. Gandhi, C. Park, V. Pham, H. Pu, S. Singh, M. Tandon, J. Zhao, H. Kim. **"Toward End-to-End ML Support on Vortex."** *Presented at the Open-Source Computer Architecture Research Workshop at ISCA 2026* **(OSCAR'26)**
- V. Venkitaraman†, M. Tandon†, T. Thorawade, K. Kokkiligadda, S. Deshmukh, A. Sathyan, N. Boran, V. Singh. **"An Energy-Efficient Virtual-Real Cache Hierarchy."** *Presented at the IEEE International System-on-Chip Conference* **(SOCC'26)**
- V. Venkitaraman†, M. Tandon†, T. Thorawade, N. Barhate, K. Kokkiligadda, N. Singh, R. Choudhary, V. Singh. **"RAGE: A Data-Aware Aggressive Prefetcher for High-Performance Graph Analytics."** *Presented at the IEEE Computer Society Annual Symposium on VLSI* **(ISVLSI'26)**
- V. Venkitaraman, T. Thorawade, M. Tandon, K. Kokkiligadda, V. Singh, J. Patel. **"LiC: Low-Cost Cache Replacement Algorithm for All Cache Levels."** *Presented at the 33rd IFIP/IEEE International Conference on Very Large Scale Integration* **(VLSI-SoC'25)**

† *denotes equal contribution*

Research Experience

CPU Architecture for Efficient LLM Inference *(Oct 2025 - Present)*
Prof. Jongse Park, KAIST

- Profiled LLM serving across **retrieval-augmented** and **agentic** pipelines and found the CPU under **5%** utilized and active only in short **bursts**, an opportunity to raise its utilization by handing it part of the inference work
- Characterized **Llama3-8B** and **Qwen2-7B** on an **Intel Xeon Gold** CPU, where a bimodal predictor came within 1% IPC of **TAGE** and prefill throughput held with the reorder buffer at **38%** of its size, all pointing to over-provisioned hardware
- Found the caches and hardware prefetchers capturing little of the weight and **KV-cache** traffic, and proposed a CPU mode for LLM inference that trims this over-provisioned hardware and serves that traffic more efficiently
- Working on an **RVV**-enabled RISC-V model in **Sniper**, calibrated against the **XiangShan** core, to evaluate the performance and energy of the proposal against a general-purpose baseline

Actor-Based Execution on GPUs *(May 2026 - Present)*
Prof. Hyesoon Kim, Georgia Institute of Technology

- Implemented the actor model on **Vortex**, an open-source RISC-V GPU, mapping one actor per hardware thread, and found global-memory message passing to consume **97%** of cycles, causing a 9× slowdown against the non-actor baseline
- Traced the cost to scattered writes the **coalescer** cannot merge, leaving one DRAM write per message sent, and to the one-actor-per-thread mapping, which fixed the actor count to the machine size and left the GPU issuing at **6%** of peak
- Proposed a **message transport** that appends each message with one **L2 atomic**, groups them by **destination** so a warp's writes land together, sizes each group to fit the receiver's cache, and drains one receiver's queue per warp
- Achieved 35.8× and 25.6× speedups on **Histogram** and **Index Gather** over naive actor versions and 6.3× and 1.5× over their non-actor versions, and these gains grow as the problem scales
- Working on **hardware/software co-design** of lightweight send and receive instructions, and on generalizing the layer to other workloads dominated by fine-grained scattered communication

Statically Bound Dataflow for Sparse Matrix Multiplication *(Jun 2026 - Sep 2026)*

Independent Research, IIT Bombay

- Studied **HiT**, the state-of-the-art sparse matrix multiplication accelerator, and found its multipliers holding **25%** of chip area and **37%** of power while the units that pair and gather their operands hold **42%** and **52%**
- Traced this to **intersection**, a comparator array searching every cycle for nonzeros to share a lane and matching **0.12%** of the time, and **transport**, a reduction so widely spread that nearly every partial sum rides a ring to its accumulator
- Realized the pattern is known offline, and proposed a dataflow that **binds** the schedule by longest-processing-time **list scheduling**, fixing which nonzero fills which lane each cycle, and **folds** a cluster's 32-cell ring into a 4 by 8 grid
- Showed binding removes the comparator array, since the lanes now follow a fixed order rather than a search, and folding removes the ring and half the buffer ports, since partial sums now walk an 8-cell chain
- Modelled the design in a cycle-accurate simulator on **SuiteSparse** matrices and measured 3.08× over HiT on every sparse-sparse workload and 1.37× across the rest, on a 17.9% smaller chip drawing 19.4% less power in the sparse regime

Efficient LUT-based Processing-using-Memory Architecture *(Jan 2026 - Jun 2026)*

Prof. Virendra Singh, IIT Bombay

- Characterized **15 workloads** across cryptography, signal processing, bioinformatics and machine learning on the state-of-the-art **pLUTo** architecture, and observed that **LUT loading** and **row sweeping** consume **95%** of execution time
- Traced the overhead to two design limits, a subarray holding only one LUT while **11 of 15** workloads require over 20 LUTs, and a full row sweep on every query even though more than half of all LUT entries go unused in **9 of 15** workloads
- Proposed **PM**, addressing both with **Packing**, which keeps multiple LUTs co-resident in a subarray to avoid reloads on function switches, and **Masking**, which skips row activations for LUT indices absent from the current input
- Estimated area, latency, and energy by extending the **pLUTo simulator**, synthesizing the mask-generation logic in **Yosys**, and modelling the broadcast network in **CACTI 7** at 22nm, with circuit-level functionality verified in **ngspice**
- Demonstrated a 4.3× speedup, 7.4× energy reduction, and 46.5× EDP improvement over pLUTo at 14.2% area overhead, alongside 86.7× and 14.3× speedups over an **Intel Core i9-14900** CPU and an **NVIDIA RTX 4070** GPU

Memory Hierarchy Design *(Nov 2024 - Mar 2026)*

Prof. Virendra Singh, IIT Bombay

1. **Virtual-Real Cache Hierarchy**
   - Studied the **TLB** energy of **VIPT** L1 caches, which translate on every access and draw up to 13% of core power, and found **VIVT** caches remove those lookups only to move the cost onto the L2 critical path as synonym handling
   - Proposed a two-level hierarchy with a VIVT L1 backed by **CBIT**, a **location table** built from the L2 tag array
   - Resolved each L1 miss by indexing CBIT with the translated physical address, whose **valid and level bits** flag the block as a synonym in L1, resident in L2, or absent, and whose **way field** gives a **tag-free read** from either cache
   - Modelled the proposal in **ChampSim** and simulated it on **49 SPEC CPU2006 and CPU2017** traces as well as multi-threaded **CloudSuite** workloads in a quad-core setup, with energy and area derived from **CACTI 6.5** at 32nm
   - Improved performance by 8% over a VIPT baseline and 15% over prior-art **VC-DSR**, while cutting dynamic cache energy by up to 64% and replacing the 0.57mm^2 L2 tag array with a 0.21mm^2 CBIT
2. **Data-Aware Prefetching for Graph Workloads**
   - Profiled the **GAP** kernels by **data type** and observed low-reuse structure blocks evicting higher-utility property data hierarchy-wide, limiting the L2 to a **16%** average hit rate, low enough that removing it improved performance
   - Traced this to **pointer chasing** through the **CSR** offset, structure, and property arrays, whose data-dependent load-load chains stride prefetchers cannot follow and recent prefetchers like **Berti** and **Pythia** cannot learn
   - Proposed **RAGE**, which isolates structure data in a **structure cache** replacing the L2 and reserves L1 and LLC for property data, generating **dependent property prefetches** at the memory controller from structure prefetch returns
   - Modelled RAGE on **Sniper** and simulated a 4-core system across five GAP kernels and eight graphs exceeding LLC capacity, with access latencies derived in **CACTI** and chip area in **McPAT** at 45nm
   - Achieved a 24% average speedup over no prefetching, outperforming recent general-purpose prefetchers and leading the closest hardware graph prefetchers **DROPLET** and **SDC+LP** by 5% and 7%, at 0.03% chip area overhead
3. **Lightweight Cache Replacement**
   - Studied the cost of **LRU**, whose recency state scales with associativity, and of LLC-based policies such as **SHiP** and **Hawkeye**, which add large predictor tables, leaving no low-overhead policy usable at all cache levels
   - Proposed **LiC**, which reframes the problem from deciding which block to evict to deciding which block to protect, retaining the most recently accessed block of a set with a single $\log_2 n$-bit pointer per set in an n-way cache
   - Modelled LiC on **ChampSim** and simulated it on **49 SPEC CPU2006 and CPU2017** single-core traces as well as multi-threaded **CloudSuite** workloads in a quad-core setup, with its storage and power estimated in **CACTI** at 45nm
   - Matched LRU's performance to within 0.001% while shrinking the replacement-policy storage of a 2MB 16-way LLC by 16×, 14×, and 28× against LRU, SHiP, and Hawkeye, and by 4× against low-cost policies **NRU** and **PLRU**

Awards & Achievements

- Awarded the **Global Korea Scholarship (GKS)** by the Korean Government for a semester exchange at KAIST *(2025)*
- Awarded a **Change of Department** to Electrical Engineering in recognition of outstanding academic performance *(2023)*
- Ranked in the **top 1.2%** among 160,000+ candidates in the Joint Entrance Examination (JEE) Advanced *(2022)*
- Ranked in the **top 1.1%** among 1,000,000+ candidates in the Joint Entrance Examination (JEE) Main *(2022)*

Teaching & Mentorship Experience

Institute Student Mentor *(Aug 2026 - Present)*

Student Mentorship Program, IIT Bombay

- Selected from **340 applicants** through rigorous process of Statement of Purpose screening, peer reviews, and interviews
- Entrusted with personally mentoring **12 freshmen** through a smooth transition to IIT Bombay, acting as their first point of contact for academics, course planning, extra-curricular activities, and personal difficulties

Teaching Assistant

Responsible for invigilating tests, grading, assisting with **course evaluation**, preparing **quiz solutions**, and mentoring students through **doubt-solving sessions**.

- EE 325: Probability and Random Processes (**130+ students**) | *Prof. Nikhil Karamchandani* *(Aug 2026 - Present)*
- EE 739: Processor Design (**170+ students**) | *Prof. Virendra Singh* *(Jan 2026 - Apr 2026)*
- EE 224: Digital Systems (**95+ students**) | *Prof. Virendra Singh* *(Aug 2024 - Nov 2024)*

Academic Service

- **Artifact Evaluation Committee Member**, *MICRO 2026*
- **Artifact Evaluation Committee Member**, *PACT 2026*

Technical Skills

Languages	C, C++, Python, VHDL, Verilog, Assembly (8051)
Software	Git, Yosys, OpenSTA, Intel Quartus, Fusion360, MATLAB, KiCad
Simulators	ChampSim, Sniper, gem5, Vortex, CACTI, ngspice
Profiling & Analysis	perf, Intel VTune, NVIDIA Nsight

Relevant Coursework

Computer Architecture & Systems: Processor Design, Systems for Machine Learning, Embedded Systems Design, Microprocessors, Microprocessors Laboratory

Digital Design: VLSI Design*, Digital Systems, Digital Circuits Lab, Algorithmic Design of Digital Systems, Electronic Design Lab *(Advanced Digital System Design)*†

Computer Science & Machine Learning: Introduction to Data Science and Machine Learning, Programming for Data Science, Discrete Structures, Computer Programming and Utilization

Mathematics: Probability and Random Processes, Linear Algebra, Calculus I & II, Differential Equations, Introduction to Optimization Techniques†

† *KAIST exchange courses* * *ongoing*